



Society of Petroleum Engineers

SPE-227232-MS

Evaluating the Effect of Friction Reducers in Slickwater on Shearing Properties Between Fracture Surfaces of Shale Rocks

MahammadShayban Boda and Hasan J Khan, College of Petroleum Engineering and Geoscience, King Fahd University of Petroleum & Minerals, KSA; Murtadha J. AlTammar, EXPEC Advanced Research Center, Saudi Aramco, KSA; Abdul Samad Mohammed and Mirza Murtuza Baig, Department of Mechanical Engineering, King Fahd University of Petroleum & Minerals, KSA; Abeer A. Alarawi, EXPEC Advanced Research Center, Saudi Aramco, KSA; Ruud Weijermaars and Theis Solling, College of Petroleum Engineering and Geoscience, King Fahd University of Petroleum & Minerals, KSA

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227232-MS](https://doi.org/10.2118/227232-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Hydraulic fracturing operations involve complex interactions among fracturing fluids, proppants, and rock surfaces, where tribological factors like friction, lubrication and wear play an important role in fracture propagation and proppant transport. This study focuses on examining the effect of friction reducers on the coefficient of friction (COF) during interaction of two shale rock surfaces. A calcareous shale block is cut orthogonal to the bedding plane to extract 5.5mm D × 15mm L pins and 37.5mm D × 15mm L core samples. Both the surfaces are polished to achieve a smooth uniform finish. A pin-on-disk tribometer is used to measure the coefficient of friction (COF) and study the frictional behavior of shales at a constant load of 15 N, with varying sliding speeds and rock/fluid combinations (dry, water environment, and slickwater with friction reducer). Pre- and post-test surfaces were analyzed using SEM, μ XRF mapping, and surface roughness measurements to assess changes in the rock surface. Preliminary results for dry shale samples show that the COF ranged between 0.75 and 0.90, with the COF decreasing with an increase in the sliding speed. This is potentially due to the frictional heat generated during the process caused by dehydration of hydrous clay minerals in the shale, releasing water that formed a lubricating layer over the contact area. Additionally, localized frictional heating may have led to the formation of a frictional melt on the shale surface, further decreasing COF during sliding. In tests with shale saturated in DI water, the COF initially ranged from 0.2 to 0.4, indicating the lubricating effect of water, which prevented direct rock-to-rock contact. Over time, as the water evaporated, the COF increases to the values comparable to the dry state, reflecting the transition back to dry conditions. In tests conducted with slickwater containing friction reducer, the COF remained relatively low and stable across all sliding speeds, converging near 0.68–0.70. This consistent performance is attributed to the formation of a persistent boundary film by the friction reducer, which effectively minimized interfacial shear and reduced sensitivity to sliding velocity. In this work, we replicate the sliding behavior of rock surfaces during the hydraulic fracturing process using the

pins and rock from the same shale samples in a pin-on-disk tribometer. The study provides novel insights about the effect of slickwater fracturing fluid on rock shearing in shale.

Introduction

Unconventional hydrocarbon resources, such as shale gas and tight oil, have become pivotal in addressing the world's increasing energy demands. These resources are trapped in low-permeability rocks (50-2000 nano-Darcy), making their extraction particularly challenging [1]. As a result, hydraulic fracturing (HF) has been essential in enhancing reservoir conductivity by creating artificial fracture through high-pressure fluid injections. Hydraulic fracturing allows trapped hydrocarbon from matrix to flow towards the wellbore for production [2]. However, the success of hydraulic fracture does not solely depend on the fracture creation, rather it involves complex interplay of mechanical, hydraulic and tribological factors at the rock-fluid and rock-rock interfaces that determine fracture propagation, stability and post-fracture conductivity [3].

In the context of hydraulic fracturing, tribology - the study of friction, wear and lubrication - plays a crucial role in controlling fracture behavior and long-term well productivity. As fracture propagates and closes, rock to rock contact occurs along shear planes and bedding interfaces. These contacts experience substantial stress and shear displacement, making their frictional characteristics a key factor in fractures propagation and proppants transport in the fracture [4], [5]. Additionally, frictional interaction between shale rock surfaces may influence the fault reactivation and induce seismicity particularly in shear slippage prone formations [6].

One of the major operations challenged in hydraulic fracturing is high friction pressure losses as fluids are injected at extremely high rates through the long wellbore and narrow perforation zone. This pressure loss increases energy demand and limits the achievable injection rate during operation which directly impacts fracture propagation and operation efficiency [2,7]. To address this problem, friction reducer (FR) typically high molecular weight chain polymer is added to fracturing fluid to prevent turbulence flow and drag reduction in wellbore. This enables higher fluid injection rate without exceeding surface pressure constraints, thereby improving the operational efficiency and effectiveness of hydraulic fracturing operations [8,9,10].

While the primary function of friction reducer is hydraulics, some studies suggest that they may also impact the tribological behavior at the rock interface. Upon shale surface contact, friction reducer molecules can absorb on the rock surfaces forming a lubrication film also known as tribofilm that reduces the real contact area and interfacial shear stress during rock-rock sliding [11]. This could reduce the coefficient of friction (COF) and alter the shale surface behavior of fracture. However, the chemical stability and evolution of these fluid films under stress and frictional heating are not well understood [12]. Investigating how friction reducers influence shale-on-shale interaction is essential to understand fracture slip during hydraulic fracturing and fault slip potential in unconventional reservoirs [13,14].

Several authors have examined the tribological aspects of hydraulic fracturing [3,13,15]. For instance, Xiao et al. (2016) conducted experiments using a ball-on-disk tribometer to evaluate the frictional properties of shale rock and proppants in the presence of water and various treatment fluids, under different load conditions and sliding velocities [4]. Their results show that the coefficient of friction (COF) is less for wet rock samples compared to dry rock samples. This implies that the de-ionized (DI) water forms a tribofilm on the rock surface and reduces the friction between rock-rock surfaces. In simple terms, smoother interactions mean more effective fracturing, better fracture networks and higher hydrocarbon recovery [5,16].

In a related study, Zhang et al. (2019) showed that guar gum aqueous solutions also reduce the COF compared to DI water. However, they observed that the degree of friction reduction is strongly dependent on rock mineralogy and surface microhardness, indicating that the tribological performance of fluids is not universal but instead highly sensitive to shale composition [17].

Today's fracturing fluids often contain a variety of chemical additives designed to optimize their behavior under extreme conditions, improving their ability to carry proppants into fractures and maximizing resource recovery [10]. Despite these valuable insights, most existing studies, including those by Xiao et al. and Zhang et al., focus on often employing simplified contact configuration (e.g., rock-on-metal or rock-on-silica). These experimental approaches fail to capture the mechanical and thermal complexities of true rock-rock shearing, where prolonged rock sliding, frictional heating and tribofilm degradation are critical factors influencing long-term frictional behavior under subsurface conditions.

This lack of experimental data about shale-on-shale sliding using field relevant fluids under extended loading conditions represents a major knowledge gap. Understanding the long-term evolution of COF, the mechanical integrity of tribofilm, and the surface transformation of shale under continuous shearing is essential for predicting fracture rock slip and fault reactivation in unconventional reservoirs.

To address the limitations of previous studies and better replicate subsurface conditions, this study investigates the long-term tribological behavior of shale-on-shale interfaces under different fluid conditions using a pin-on-disk tribometer. Both the pin and disk are fabricated from Eagle Ford shale, cut orthogonal to the bedding plane to simulate natural shearing along lamination planes. Experiments are conducted under a constant normal load of 15 N, with sliding performed in three test environments: dry condition, DI water and friction reducer. The coefficient of friction (COF) is recorded continuously throughout the test duration, capturing the transition from initial contact to steady-state sliding.

This study aims to:

1. Quantify how friction reducers influence COF under extended shear displacement.
2. Compare long-term frictional trends across three test environments: dry, DI water, and Friction reducer (FR).

The findings of this study help evaluate how tribological evolution impacts key geomechanical phenomena such as fracture slip behavior, proppant embedment resistance and fault reactivation potential. Lower COF values, particularly in the presence of friction reducers, may facilitate fracture propagation and reduce shear resistance but can also increase the risk of fault slip or induced seismicity, especially in formations containing critically stressed planes. A deeper understanding of these tribological interactions is essential for designing safer and more effective hydraulic fracturing treatments, optimizing fluid formulations, and mitigating geomechanical risks associated with fluid injection into tight or unstable reservoir.

Materials and methods

Rock Sample

The Eagle Ford shale sample used in this study is categorized by low permeability mainly in the nano-Darcy range; this is representative of tight reservoir rock [[18]. X-ray diffraction (XRD) analysis confirmed that the sample is predominantly composed of calcite (87.2%) with quartz (12.8%) indicating carbonate rich mineralogy. The total organic carbon (TOC) content is measured as 2.19% by Rock-Eval pyrolysis.

Rock Sample Preparation

As shown in [Figure 1\(a\)](#), the rock disks were prepared by cutting a 1-inch diameter core from Eagle Ford shale, with the cutting orientation maintained perpendicular to the bedding planes to simulate natural shear along laminations. The disks were then sectioned into final dimensions of 37.5 mm in diameter and 15 mm in thickness.

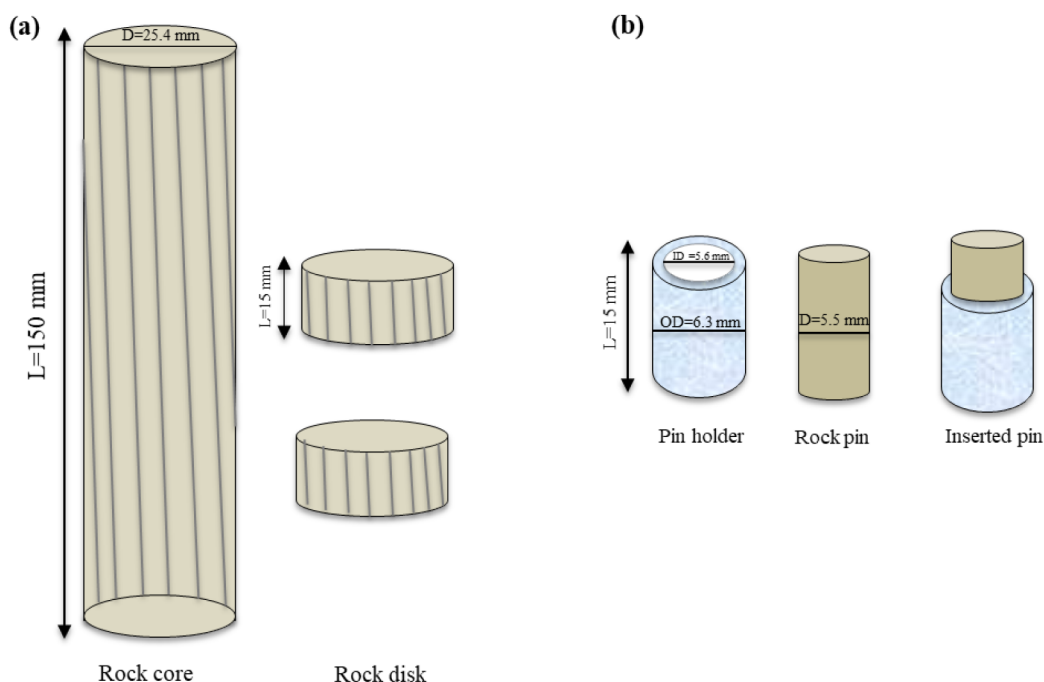


Figure 1—Schematic of shale rock sample preparation. (a) Rock disks; (b) Rock pin.

A rock pin with a 6.3 mm outer diameter (OD) was required for the pin-on-disk tribometer. Due to equipment constraints, the pin was initially drilled using a 5.5 mm inner diameter (ID) bit, yielding a rock pin of 5.5 mm diameter and 15 mm length. To ensure compatibility with the tribometer, the rock pin was then inserted into a custom-designed 3D-printed holder with an outer diameter of 6.3 mm (Figure 1(b)).

Both the rock disk and pin surfaces were polished in two stages, beginning with 120-grit sandpaper for initial leveling, followed by 600-grit sandpaper to achieve a smooth and consistent contact area suitable for accurate and repeatable tribological measurements.

Test fluids

Two fluid environments were used to investigate the tribological response of shale surfaces: deionized (DI) water and slickwater containing friction reducers.

Deionized water served as the baseline fluid, enabling evaluation of basic lubrication effects without the influence of chemical additives. This allowed for the study of tribofilm formation due to water-rock interactions alone. To simulate realistic fracturing fluid, a commercial slickwater formulation was prepared in DI water that included a commercial friction reducer, surfactant, biocide, and clay stabilizer. The measured rheological properties of slickwater fluid exhibited a density of 1.004 g/ml, a pH of 6.8 and a viscosity of 2.07 cP at room temperature and pressure conditions.

Pin-on-disk tribometer

A tribometer is an instrument used to measure the coefficient of friction (COF) and to evaluate the frictional behavior of materials under control contact conditions. Among the various configurations available, the pin-on-disk tribometer is widely used due to its simplicity and reliability. In this study, the pin-on-disk configuration was selected because it offers an effective and repeatable method to evaluate the lubrication performance of different fluids under sliding contact conditions. According to ASTM G99 standards, the test involves a stationary rock pin (the primary specimen) that is loaded vertically (F_n) against a rotating disk (the counter face). The disk rotates at a constant speed and the frictional force (F_f) generated at the contact interface is continuously measured enabling precise calculation of COF throughout the test duration. The schematic illustration and actual instrument setup are shown in Figure 2.

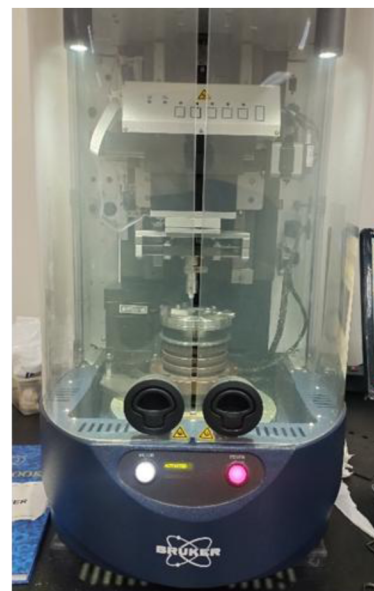
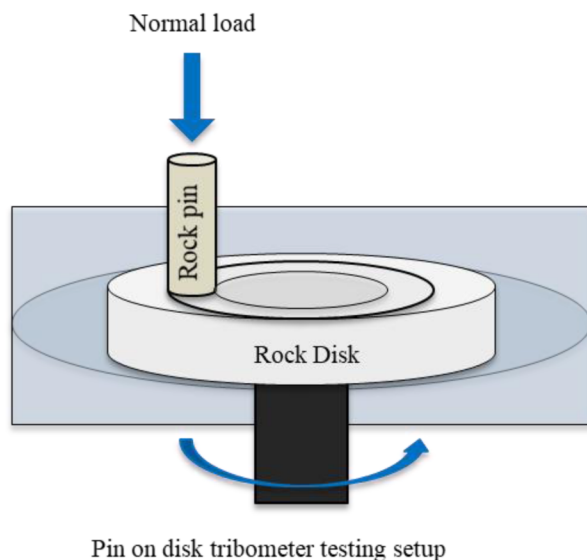


Figure 2—(Left) Schematic illustration of the pin-on-disk tribological testing setup, showing the vertical loading of a shale pin onto a rotating rock disk. (Right) Photograph of the Bruker MMT-3MT tribometer used in this study.

The coefficient of friction (COF) was calculated using Amontons' Law, expressed as [19]:

$$\text{COF}(\mu) = \frac{F_r}{F_n} \quad (\text{Eq. 1})$$

where F_r is the measured frictional force, and F_n is the applied normal load.

Testing procedures

Disk and pin specimens were prepared from the Eagleford shale samples and polished to ensure consistent contact surfaces, as detailed in Section 3.2.

Tribological testing was performed using a pin-on-disk tribometer under a constant normal load of 15 N, with sliding speeds ranging from 10 to 50 mm/s, depending on the test condition. Each test was conducted for 1500 cycles, resulting in a total sliding distance of ~5655 cm. Surface roughness analysis was performed, using a laser confocal analyzer, before and after testing to evaluate changes in surface topography due to sliding and fluid interaction in the pin-on-disk tribometer.

Three distinct test cases were designed to isolate the effects of sample saturation and fluid lubrication during sliding. These are summarized in Table 1.

Table 1—Summary of Tribological Test Cases.

Case	Sample condition	Testing environment	Purpose
1	Dry	Dry	Baseline – pure mechanical friction
2	Dry	DI water	Lubrication from DI water
3	Dry	Slickwater fluid that includes a friction reducer (FR)	Assess friction reducer's lubrication

Results and discussion

Case 1 (Dry sample in dry testing condition)

In this case, both the rock pin and disk were tested in dry conditions without any fluid application. The tests were conducted under a constant normal load of 15 N at varying sliding speeds (10–50 mm/s).

Figure 3 presents the evolution of the coefficient of friction (COF) with sliding distance for the dry tests. Across all velocities, the COF showed a progressive increase with displacement. Initial values were observed in the range of 0.1 to 0.2, followed by a velocity-dependent rise. At 10 mm/s, the COF sharply increased, exceeding 1.0 before reaching a steady state. In contrast, tests at 30 and 50 mm/s displayed more gradual COF development and ultimately stabilized at lower values in the range of 0.5–0.6

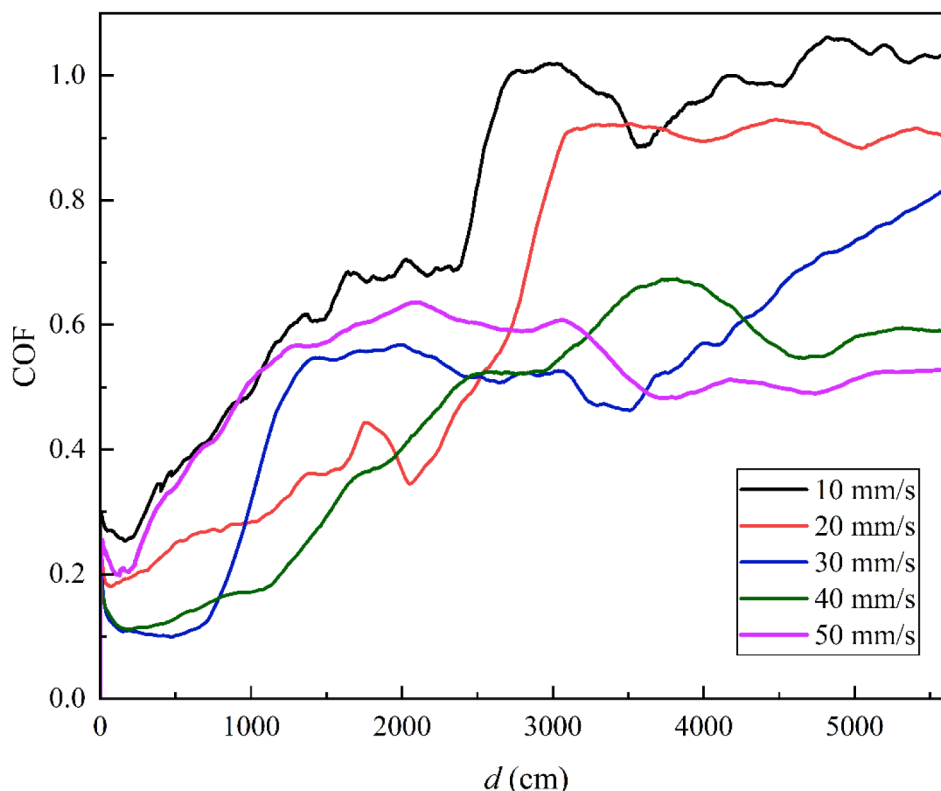


Figure 3—Coefficient of friction (COF) versus sliding distance (d) for case 1 under different sliding velocities (10 – 50 mm/s).

This behavior is likely influenced by debris generation at the contact interface. Under dry sliding conditions, wear particles may accumulate between the contacting surfaces and function as third-body elements, promoting rolling or sliding between the rock surfaces. This three-body mechanism can decrease direct asperity engagement and may explain the lower steady-state friction observed at higher sliding velocities, where debris dispersion is more efficient and contact surfaces undergo smoother conditioning.

The COF evolution in all tests followed a typical three-stage trend: an initial run-in phase marked by asperity accommodation, a transitional phase characterized by increasing friction due to growing real contact area, and a steady-state regime where COF plateaued with continued sliding [16,17]. The influence of sliding velocity was evident: higher speeds led to earlier stabilization and lower friction, likely due to accelerated surface adaptation and diminished abrasive interaction. These results provide a baseline for subsequent comparisons with fluid-assisted lubrication environments.

Case 2 (Dry sample in wet testing condition (DI water))

In this test condition, the dry samples were fully submerged in DI water throughout the test duration to enable real-time fluid-assisted lubrication. The tests were conducted under a constant normal load of 15 N, at sliding speeds of 10, 30 and 50 mm/s. The evolution of the coefficient of friction (COF) as a function of sliding distance is presented in Figure 4.

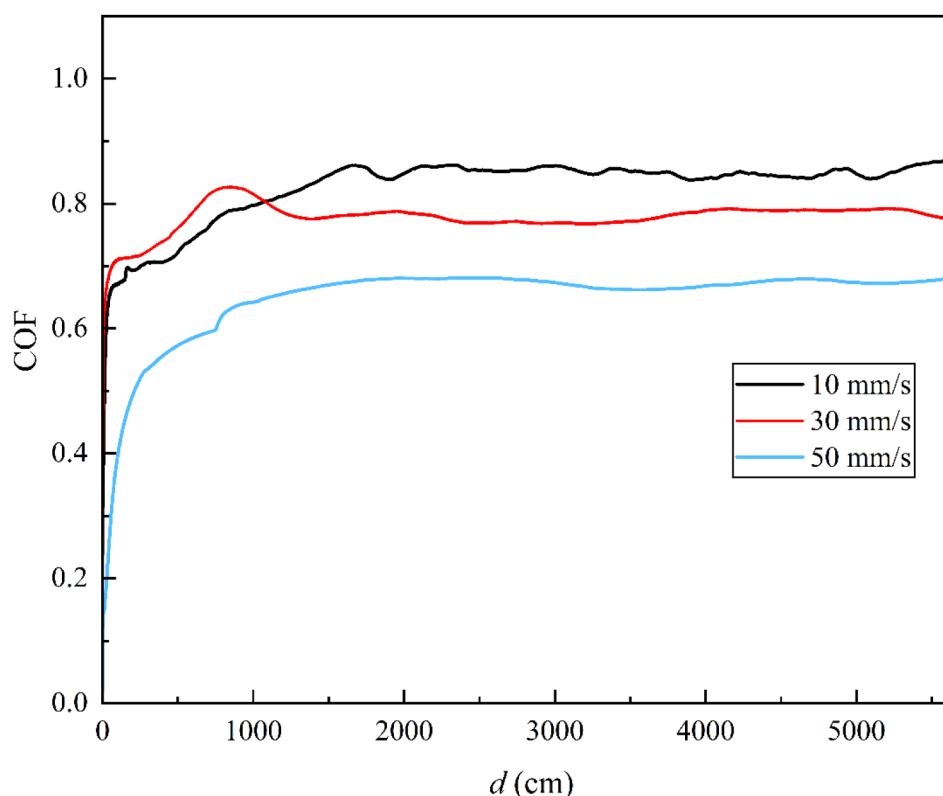


Figure 4—Coefficient of friction (COF) versus sliding distance (d) for case 2 under three sliding velocities (10, 30 & 50 mm/s).

Across all speeds, the coefficient of friction (COF) initially ranged from ~ 0.2 to 0.4 , increasing gradually before stabilizing into a steady-state regime. Notably, the steady-state coefficient of friction (COF) decreased with increasing sliding speed: the test at 10 mm/s stabilized at approximately 0.83 , while the test at 50 mm/s reached a lower value of ~ 0.63 . The 30 mm/s condition exhibited intermediate behavior, with coefficient of friction (COF) stabilizing near 0.70 . This inverse relationship between sliding velocity and steady-state coefficient of friction (COF) suggests that hydrodynamic or boundary lubrication mechanisms are more effective at higher velocities, potentially due to enhanced fluid film thickness in-between sliding rock surfaces [20].

The DI water environment led to more stable COF behavior compared to dry conditions, with notably lower friction at lower sliding speeds (10 and 30 mm/s). However, at 50 mm/s, the dry condition exhibited a slightly lower steady-state COF, potentially due to third-body interactions caused by debris accumulation that reduced direct asperity contact. The presence of the water medium likely suppressed direct asperity contact, minimized abrasive interactions, and allowed for smoother interfacial adaptation [15].

Case 3 (Dry sample in wet testing conditions (Slickwater fluid containing friction reducer))

This case evaluates the tribological response of dry shale samples tested under continuous lubrication with slickwater fluid that contains a friction reducer. Tests were conducted under a constant normal load of 15 N, at sliding speeds of 10 , 30 , and 50 mm/s, with the samples fully immersed in the FR fluid during sliding. Figure 5 presents the evolution of the coefficient of friction (COF) as a function of sliding distance.

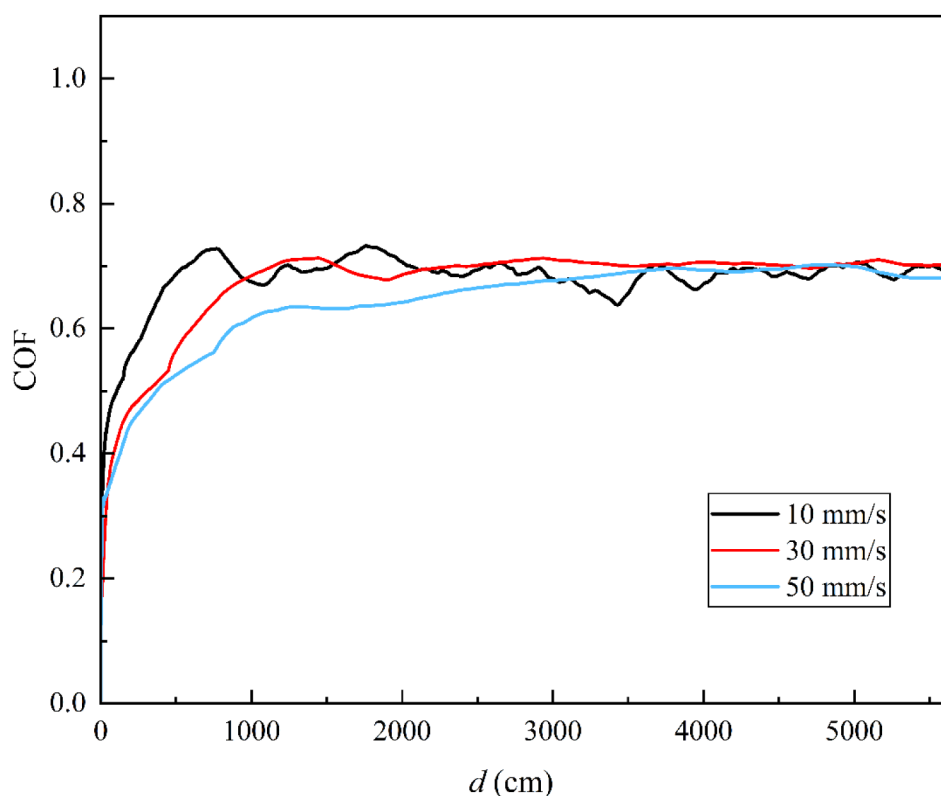


Figure 5—Coefficient of friction (COF) versus sliding distance (d) for case 3 under three sliding velocities (10, 30 & 50 mm/s).

Across all test conditions, the COF exhibited relatively low and stable values compared to all other cases. The initial COF values ranged from ~ 0.3 to 0.5 , and steady-state values converged toward ~ 0.68 – 0.70 regardless of sliding speed. Although the 10 mm/s condition showed slightly higher fluctuation during the transition phase, all speeds exhibited rapid stabilization, and the differences in final COF across speeds were minimal.

The reduced and stabilized friction response indicates the efficacy of FR additives in forming persistent boundary lubrication films on the shale surface. These polymers and surfactants likely contributed to shear-thinning behavior and interfacial film formation that limited asperity interaction. The narrow COF range across different sliding velocities suggests that the FR system provided robust lubrication performance, less sensitive to speed-dependent fluid film breakdown.

Compared to the DI water environment (Case 2), the slickwater produced a more consistent and slightly lower COF profile, indicating improved lubrication characteristics. These findings support the application of polymer-based friction reducers as effective interfacial modifiers in hydraulic fracturing, reducing sliding resistance at rock interfaces even under extended displacement.

Comparison of all cases

Figure 6 represents a comparative summary of steady-state coefficient of friction (COF) as a function of sliding velocity (v) for the three test cases. Across all cases, an overall decreasing trend in COF with increasing sliding velocity is observed, though the extent of this effect varies by test condition.

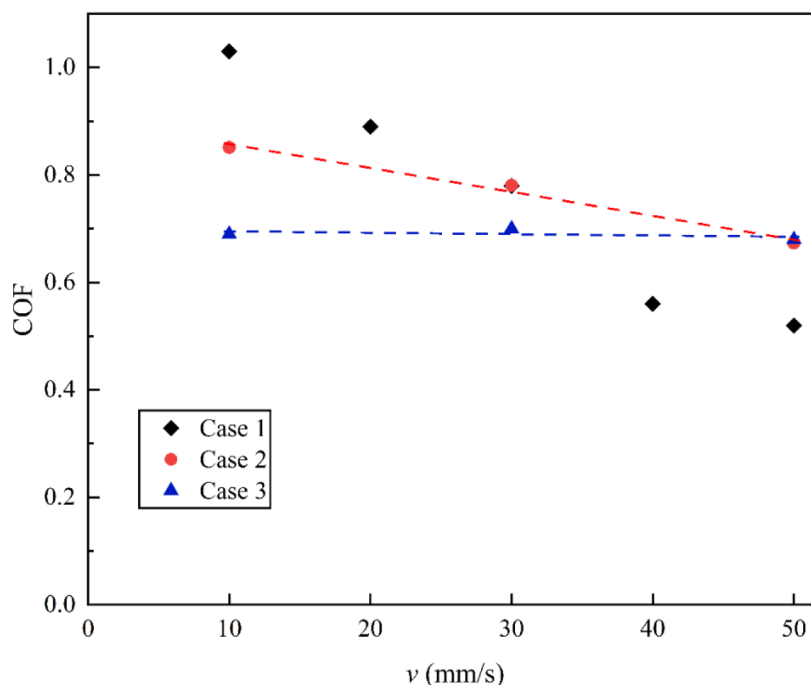


Figure 6—Comparison of steady-state coefficient of friction (COF) as a function of sliding velocity (v) for all cases.

Case 1 (dry condition) shows the highest sensitivity to velocity, with a steep reduction in COF from over 1.0 at 10 mm/s to below 0.6 at 50 mm/s. This suggests that dry sliding is strongly influenced by abrasive contact and debris accumulation, which are more effectively mitigated at higher speeds. Case 2 (DI water) also exhibits a declining trend, though less pronounced, indicating partial hydrodynamic lubrication at increased velocities. In contrast, Case 3 (Slickwater with FR) demonstrates the most stable performance, with minimal variation in COF across all tested velocities, confirming the velocity-independent lubrication behavior of polymeric friction reducers.

This comparative analysis emphasizes the importance of continuous fluid presence and the superior performance of friction reducers in maintaining low and consistent friction under varied shear conditions.

Conclusion

This study examined the tribological behavior of shale-on-shale contact under varying fluid environments and sliding velocities using a pin-on-disk tribometer. The test matrix included three key scenarios: dry contact (Case 1), lubrication with DI water (Case 2) and lubrication using a polymer-based friction reducer (FR) in slickwater (Case 3). The results highlight the critical role of interfacial fluid dynamics and sliding velocity on the evolution of the coefficient of friction (COF) during extended displacement, with direct implications for hydraulic fracturing operations and subsurface slip modeling.

- Case 1 (Dry sample, dry environment): Exhibited the highest frictional resistance, with COF values exceeding 1.0 at lower velocities and stabilizing between 0.8–0.9 in steady-state. The frictional response followed a typical tribological sequence: run-in, transitional increase due to asperity engagement, and stabilization. Under dry conditions, progressive wear generated debris that contributed to three-body interactions, influencing the sliding contact by intermittently reducing direct asperity engagement. At higher velocities, improved debris dispersal may have led to smoother contact and lower COF. These dry test results establish the baseline for evaluating fluid-mediated frictional behavior.
- Case 2 (Dry sample, DI water environment): Demonstrated significant frictional reduction relative to the dry case, with steady-state COF ranging from ~0.63 to 0.83, depending on sliding speed.

The DI water provided effective real-time lubrication, reducing shear resistance by maintaining a fluid layer at the contact interface. The inverse relationship between velocity and COF suggests that hydrodynamic entrainment may occur at higher speeds, enhancing fluid thickness and limiting direct asperity interaction.

- Case 3 (Dry sample, Slickwater containing FR): Delivered the most stable and velocity-independent performance, with steady-state COF converging near 0.68–0.70 across all sliding speeds. The polymeric friction reducer formed persistent boundary films and exhibited shear-thinning behavior, enabling consistent lubrication and minimal speed sensitivity. These results validate the superior lubricating properties of FR-treated fluids in shale interfaces, especially under prolonged shear displacement.

Overall, the findings underscore that maintaining real-time fluid presence at the sliding interface is critical for effective friction control in shale-on-shale interface. While both DI water and FR in slickwater provided lubrication, the FR delivered more consistent and robust performance across all tested velocities. The influence of sliding velocity was more pronounced under dry and water-lubricated conditions but was significantly mitigated in the presence of polymeric friction reducers due to their stable film formation and shear-thinning behavior.

These insights are directly applicable to the optimization of slickwater fracturing designs and the modeling of interfacial slip behavior along bedding planes or faulted shale layers. Future work should explore high-temperature effects, fluid degradation mechanisms, and computational modeling to better simulate in-situ hydraulic fracturing conditions.

References

1. J. Speight, 'Reservoirs and reservoir fluids', *Shale Oil and Gas Production Processes*, pp. 139–220, 2020, doi: [10.1016/B978-0-12-813315-6.00003-8](https://doi.org/10.1016/B978-0-12-813315-6.00003-8).
2. H. Ben Mahmud et al, 'A Review of Fracturing Technologies Utilized in Shale Gas Resources', *Emerging Technologies in Hydraulic Fracturing and Gas Flow Modelling*, May 2020, doi: [10.5772/INTECHOPEN.92366](https://doi.org/10.5772/INTECHOPEN.92366).
3. G. Swan, 'Tribology And The Characterisation Of Rock Joints', Jun. 29, 1981, *OnePetro*. Accessed: Sep. 15, 2024. [Online]. Available: <https://dx.doi.org/>
4. H. Xiao, S. Liu, and D. Wang, 'Tribological properties of sliding shale rock–alumina contact in hydraulic fracturing', *Tribol Lett*, vol. **62**, no. 2, pp. 1–9, May 2016, doi: [10.1007/s11249-016-0667-x](https://doi.org/10.1007/s11249-016-0667-x).
5. Y. Guo et al, 'Friction Challenge in Hydraulic Fracturing', *Lubricants* 2022, Vol. 10, Page 14, vol. **10**, no. 2, p. 14, Jan. 2022, doi: [10.3390/LUBRICANTS10020014](https://doi.org/10.3390/LUBRICANTS10020014).
6. R. J. Davies et al, 'Oil and gas wells and their integrity: Implications for shale and unconventional resource exploitation', *Mar Pet Geol*, vol. **56**, pp. 239–254, 2014, doi: [10.1016/j.marpetgeo.2014.03.001](https://doi.org/10.1016/j.marpetgeo.2014.03.001).
7. Q. Li, H. Xing, J. Liu, and X. Liu, 'A review on hydraulic fracturing of unconventional reservoir', *Petroleum*, vol. **1**, no. 1, pp. 8–15, Mar. 2015, doi: [10.1016/J.PETLM.2015.03.008](https://doi.org/10.1016/J.PETLM.2015.03.008).
8. C. T. Montgomery and M. B. Smith, 'Hydraulic Fracturing: History of an Enduring Technology', *Journal of Petroleum Technology*, vol. **62**, no. 12, pp. 26–40, Dec. 2010, doi: [10.2118/1210-0026-JPT](https://doi.org/10.2118/1210-0026-JPT).
9. R. Barati and J. T. Liang, 'A review of fracturing fluid systems used for hydraulic fracturing of oil and gas wells', *J Appl Polym Sci*, vol. **131**, no. 16, Aug. 2014, doi: [10.1002/APP.40735](https://doi.org/10.1002/APP.40735).

10. B. Yang, J. Zhao, J. Mao, H. Tan, Y. Zhang, and Z. Song, 'Review of friction reducers used in slickwater fracturing fluids for shale gas reservoirs', Feb. 01, 2019, Elsevier B.V. doi: [10.1016/j.jngse.2018.12.016](https://doi.org/10.1016/j.jngse.2018.12.016).
11. Y. Zhou, D. N. Leonard, W. Guo, and J. Qu, 'Understanding Tribofilm Formation Mechanisms in Ionic Liquid Lubrication', *Scientific Reports* 2017 7:1, vol. 7, no. 1, pp. 1–8, Aug. 2017, doi: [10.1038/s41598-017-09029-z](https://doi.org/10.1038/s41598-017-09029-z).
12. J. Bech, N. Bay, and M. Eriksen, 'A Study of Mechanisms of Liquid Lubrication in Metal Forming', *CIRP Annals*, vol. 47, no. 1, pp. 221–226, Jan. 1998, doi: [10.1016/S0007-8506\(07\)62822-4](https://doi.org/10.1016/S0007-8506(07)62822-4).
13. A. H. Kohli and M. D. Zoback, 'Frictional properties of shale reservoir rocks', *J Geophys Res Solid Earth*, vol. 118, no. 9, pp. 5109–5125, Sep. 2013, doi: [10.1002/JGRB.50346](https://doi.org/10.1002/JGRB.50346);WGROUP:STRING:PUBLICATION.
14. C. Cornelio, E. Spagnuolo, G. Di Toro, S. Nielsen, and M. Violay, 'Mechanical behaviour of fluid-lubricated faults', *Nat Commun*, vol. 10, no. 1, pp. 1–7, Dec. 2019, doi: [10.1038/S41467-019-09293-9](https://doi.org/10.1038/S41467-019-09293-9);TECHMETA=123,128,129;SUBJMETA=2151,2809,4111,508,704;KWRD=GEOPHYSICS,NATURAL+HAZARDS,SEISMOLOGY.
15. M. Violay et al, 'Effect of water on the frictional behavior of cohesive rocks during earthquakes', *Geology*, vol. 42, no. 1, pp. 27–30, Jan. 2014, doi: [10.1130/G34916.1](https://doi.org/10.1130/G34916.1).
16. H. He, L. Luo, and K. Senetakis, 'Effect of normal load and shearing velocity on the interface friction of organic shale – Proppant simulant', *Tribol Int*, vol. 144, Apr. 2020, doi: [10.1016/j.triboint.2019.106119](https://doi.org/10.1016/j.triboint.2019.106119).
17. H. Zhang, S. Liu, and H. Xiao, 'Tribological properties of sliding quartz sand particle and shale rock contact under water and guar gum aqueous solution in hydraulic fracturing', *Tribol Int*, vol. 129, pp. 416–426, Jan. 2019, doi: [10.1016/j.triboint.2018.08.043](https://doi.org/10.1016/j.triboint.2018.08.043).
18. S. C. Hsu and P. P. Nelson, 'Characterization of eagle ford shale', *Eng Geol*, vol. 67, no. 1–2, pp. 169–183, Jan. 2002, doi: [10.1016/S0013-7952\(02\)00151-5](https://doi.org/10.1016/S0013-7952(02)00151-5).
19. B. Bhushan, 'Introduction to Tribology, 2nd Edition | Wiley', 2002, Accessed: Apr. 20, 2025. [Online]. Available: <https://www.wiley.com/en-gb/Introduction+to+Tribology%2C+2nd+Edition-p-9781119944539>
20. E. E. Brodsky and H. Kanamori, 'Elastohydrodynamic lubrication of faults', *J Geophys Res Solid Earth*, vol. 106, no. B8, pp. 16357–16374, Aug. 2001, doi: [10.1029/2001JB000430](https://doi.org/10.1029/2001JB000430).